

WELL COMPLETION REPORT (SYNTHETIC)

OIL-LKW-001 - Lakwa-001

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000004
Field / District	Lakwa / Charaideo
Block	PEL-ASSAM-04
Basin	Assam-Arakan
Formation	Barail (BAR)
Spud / Completion	2002-04-11 / 2002-09-18
Measured / TVD	3210 m / 3181 m
Trajectory	Vertical; inc 4.7 deg; az 216.4 deg
Formation Interval	1969-2859 m
Lithology	Interbedded sandstone and shale
Pore Pressure Reference	3952 psi
Fracture Gradient	15.27 ppg
Mud System	Glycol-PHPA WBM
NWIS Well Risk	High (59.1/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	802	Surface
9 5/8 in casing	1861	Intermediate
7 in liner/casing	2824	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2002-07-01 1760 m	Kick	Critical	Drilling break followed by positive flow check. Mitigation: Increase mud weight within fracture-margin limits and monitor gas trend. Lesson: Treat pit gain plus gas increase as an early influx indicator.
2002-05-02 560 m	Torque Spike	High	Rotary speed fluctuated during constant pump rate. Mitigation: Use staged parameter reduction to avoid torsional oscillation. Lesson: Avoid simultaneous high WOB and high RPM in correlated interval.
2002-05-07 662 m	Cementing Issue	High	Cement density/returns trend indicated placement uncertainty. Mitigation: Use staged cementing response when returns deteriorate. Lesson: Offset loss history should inform cementing hydraulics.
2002-05-10 739 m	Mud Loss	High	Active pit volume declined during stable pump rate. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.
2002-06-04 1233 m	Pressure Spike	High	Standpipe pressure increased rapidly at nearly stable flow. Mitigation: Condition hole and confirm pressure normalization. Lesson: Use depth-matched offset hydraulic baseline for early warning.
2002-07-20 2112 m	Torque Spike	High	Rotary speed fluctuated during constant pump rate. Mitigation: Use staged parameter reduction to avoid torsional oscillation. Lesson: Avoid simultaneous high WOB and high RPM in correlated interval.
2002-08-23 2745 m	Pressure Spike	High	Standpipe pressure increased rapidly at nearly stable flow. Mitigation: Condition hole and confirm pressure normalization. Lesson: Rate-of-change in pressure is more informative than a fixed threshold.

Completion Summary

The synthetic well reached 3210 m MD in the Barail interval. The NWIS knowledge record links 19 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.